Superposition and interference

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Learning target: Add signed displacements and distinguish interference from lost energy.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 4.*
Upright. An ideal free-end reflection does not invert the displacement pulse.

02. Worked example: explain the model
Up 3 plus down 1 gives up 2 cm.

03. Read and interpret the model
First amplitudes add; second net displacement zero at perfect overlap.

04. Apply the physics
+2 cm; 0 cm.

05. Explain your reasoning
They continue with original shapes/directions; local zero displacement does not mean energy destroyed.

06. Review: connect an earlier idea *Review: Unit 5, Day 7.*
$\Delta U = +20$ J.

07. Exit check
+3 mm; partly cancel.

08. STRETCH: test the model
Opposite equal pairs such as$\pm 1, \pm 2, \pm 4$ cm; waves continue propagating in a linear medium.

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01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 4.*
First λ=5 m; second λ=7 m, using λ=v/f.

02. Worked example: explain the model
Up 3 plus down 1 gives up 2 cm.

03. Read and interpret the model
First amplitudes add; second net displacement zero at perfect overlap.

04. Apply the physics
2 A and 0; stable phase relationship/coherence.

05. Explain your reasoning
They continue with original shapes/directions; local zero displacement does not mean energy destroyed.

06. Review: connect an earlier idea *Review: Unit 5, Day 7.*
$Q=\Delta U+W_{by}=8$ J; heat enters.

07. Exit check
-3 mm; no, energy also depends on motion/spatial gradients and surrounding wave field.

08. STRETCH: test the model
8 cm in phase; 0 cm in antiphase. Result depends on phase, not amplitude alone.